

T2 Inversion & Gen-Bias Audit

Two questions on the completed 78-cell study: why drawer_stow (T2) underperforms and inverts its level ordering, and whether the data generator's success filter contaminates the measured SR curves anywhere. Evidence base: frozen eval sets joined to per-episode outcomes, retained-vs-rejected generation attempts, task/env source, and dataset metadata.

0 · A harness bug was found, fixed, and the full surface re-measured clean — 62 evals, zero failures.

The phantom-success carryover bug (previous batch's success termination re-read at $t=0$) is real and mechanically proven, but the clean re-eval shows its material damage was confined to **T2's mid-SR cells** (L1 n400: $0.23 \rightarrow 0.15$; L2 n400: $0.41 \rightarrow 0.37$); T1, T3, and near-saturated cells reproduce their published values. Two further phenomena surfaced en route: a genuine **batch-0 depression** (first batch scores far below the rest in 37/37 non-saturated cells, mean -0.21 — present with the guard active, so not a scoring artifact) and **env-index non-uniformity** (success depends on which of the 20 clones runs the episode; $p \approx 2e-6$ in T1). One original run (T2_L1_n50) measured something else entirely and is superseded. All headline tables now come from the clean surface below.

1 · T2's low SR and the $L2 > L1$ inversion are training-side, and the failure has an exact address.

On the clean surface the inversion is larger than published (L2 0.37 vs L1 0.15 at $N=400$), and diagnostic cross-evals (archived under `results/diagnostics/`) attributed it to the policy, not the eval sets — consistent with L2's harsher generation filter keeping cleaner demos. The stage funnels localize the failure: the drawer phase is nearly solved at every level, stowing is free once the object is over the open drawer, and the single bottleneck is grasping the object mid-rollout. Undertraining, eval-horizon censoring, and coverage remain ruled out.

2 · Generator filtering exists — heavily in T2 — but inflates SR nowhere.

All 15 task×level cells audited. The filter is pose-structured in T2 (yaw) and T3 (y / bearing), yet in no cell is eval success elevated where generation succeeded (no $p < 0.01$ in the contamination direction), and nearest-training-demo distance predicts success nowhere. The filter's one demonstrated downstream effect is via demo *quality*, not coverage — and it worked in T2 L2's favor. (Phantom successes dilute these episode-level tests but cannot manufacture the observed nulls.)

Finding 0 · discovered during follow-up

The batch-boundary carryover bug

The stage instrumentation added for the follow-up evals records when each episode first succeeds — and roughly half of all recorded successes had `t_success = 0`: success on the first policy step after a batch reset, physically impossible in every task. The chain of evidence, all from data already on disk:

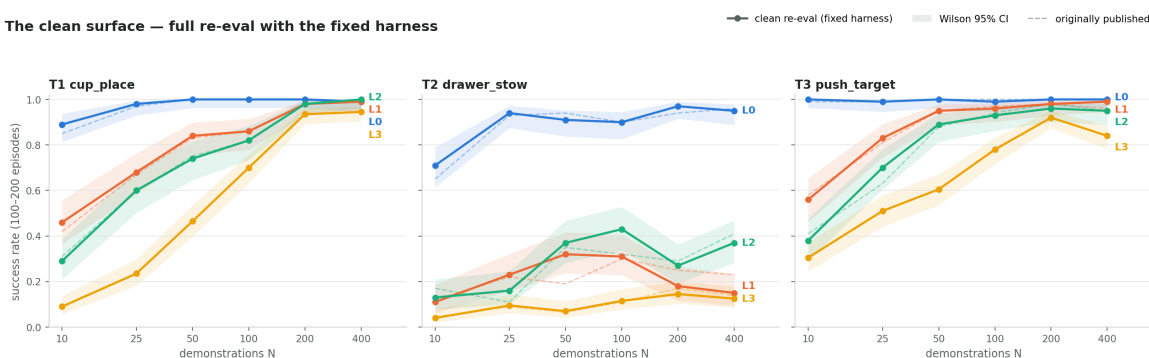
- **Zeros never occur in batch 0** (nothing to carry over): 0 of 60+ across five instrumented runs.
- **Exact carryover identity:** phantom count in batch b equals the genuine-success count in batch $b-1$ in 20 of 20 batch transitions.
- **Phantom episodes never lift the object or reach the drawer** (scene-state reads are fresh; only the termination buffer is stale, and only at $t=0$ — the earliest genuine success is step 606).
- **The published data shows the predicted signature everywhere:** batch-0 SR is the lowest batch in 36 of 38 non-saturated flat cells (sign test $p \approx 6e-10$; mean gap +0.20).

Post-hoc correction was attempted and then superseded. A constraint-propagation inversion of the carryover identity produced per-cell bounds, but its point estimates anchored on batch 0 — and the clean re-eval revealed batch 0 to be *genuinely depressed* (below), so those estimates systematically undershot and are deprecated. The user authorized a full clean re-eval instead: **54 flat cells + 6 T2 L3 cells + the D24 checkpoint pair, 62 evals, zero failures**, run on the fixed harness over ~18 h of managed local GPU time.

What the clean surface settles

- **T1 and T3 published numbers were essentially right.** Phantoms coincided with genuine successes in their near-ceiling batches; clean values match published within noise everywhere.
- **T2's mid-SR cells were genuinely inflated** (L1 n400 0.23→0.15, n200 0.25→0.18) — and one original run, T2_L1_n50 (published 0.19, clean 0.32), didn't measure today's checkpoint at all (its own hard bound caps it at 0.14; checkpoint identity re-verified via embedded train configs, task code unchanged) — most plausibly a stale weights copy in the since-deleted eval worktree.
- **Batch-0 depression is real and separate:** with the guard active, batch 0 scores −0.21 below batches 1–4 on average (37/37 non-saturated cells; T1 0.43 vs 0.75, T2 0.33 vs 0.50, T3 0.65 vs 0.85). Not a scoring bug — those episodes genuinely fail. Mechanism probe below.
- **Env-index non-uniformity:** pooled over non-saturated clean cells, success depends on env index (χ^2 p = 2.4e-6 / 1.3e-2 / 6.9e-4 for T1/T2/T3), and an env that succeeded in one batch succeeds in the next at 60–78% vs 15–32% base. The 20 clones are not exchangeable — plausibly per-clone-origin rendering differences. Protocol-level finding for any future Isaac benchmark work.
- **D24 re-check (clean): 40k/60k/80k = 0.95/0.98/0.98** — last checkpoint (joint-)best; the last-checkpoint-only protocol stands.

The clean surface — full re-eval with the fixed harness



The clean surface (solid, Wilson 95% bands) vs originally published (dashed). T1/T3: published survives. T2: L1/L2 were inflated at high N and show a genuine mid-N hump (peak near N=50–100) — with one seed, run-level variance on this task is large, so the hump's shape should not be over-read.

TASK	LEVEL	SR@10	SR@100	SR@400	N*(50%)	N*(80%)	N*(90%)	COST VS L0 @90%
cup_place	L0	0.89	1.00	0.99	≤10	≤10	11	1.0×

TASK	LEVEL	SR@10	SR@100	SR@400	N*(50%)	N*(80%)	N*(90%)	COST VS L0 @90%
cup_place	L1	0.46	0.86	0.99	12	42	126	11.4×
cup_place	L2	0.29	0.82	1.00	19	84	141	12.8×
cup_place	L3	0.09	0.70	0.94	55	134	180	16.3×
drawer_stow	L0	0.71	0.90	0.95	≤10	14	21	1.0×
drawer_stow	L1	0.11	0.31	0.15	>400	>400	>400	≥18.8×
drawer_stow	L2	0.13	0.43	0.37	>400	>400	>400	≥18.8×
drawer_stow	L3	0.04	0.12	0.12	>400	>400	>400	≥18.8×
push_target	L0	1.00	0.99	1.00	≤10	≤10	≤10	1.0×
push_target	L1	0.56	0.96	0.99	≤10	23	37	3.8×
push_target	L2	0.38	0.93	0.95	14	36	59	6.0×
push_target	L3	0.30	0.78	0.84	24	110	181	18.1×

Verdict on the study's findings, from clean data: the core claims survive — generality's demonstration cost is monotone and large in every task, and T2's cliff stands and deepens (no level above L0 reaches 50% at any N; cost ratios $\geq 18.8\times$). **Finding 4 (the object axis costs the most) does not survive scrutiny — it is confounded by the warm-up effect below.** The headline table's L3 ratios ($16.3\times/18.1\times$) compare warm flat-cell numbers against cold all-first-batch L3 numbers. Caveats that stand: one seed per cell; T2's mid-N hump (peak near $N=50-100$) is unexplained and untested against a second seed.

The batch-0 effect is process warm-up — and it re-opens the object-axis question

With the guard active, the first batch of every eval process scores far below the rest: 37/37 non-saturated clean cells, mean -0.21 (T1 0.43 vs 0.75 , T2 0.33 vs 0.50 , T3 0.65 vs 0.85). A targeted probe pinned the mechanism: a fresh process evaluating seeds 5001–5004 (skipping seed 5000 entirely) scored 0.35 on its first batch — the same seed 5001 that scores 0.85 when run warm in the main sweep. The depression follows *process position*, not *the seed*: renderer/physics warm-up. The full probe recovered batch by batch — $0.35 / 0.65 / 0.80 / 0.85$ across seeds 5001–5004, converging on those seeds' warm values ($0.85 /$

0.70 / 0.60 / 0.85) after the first batch. The warm-up costs roughly the first 20 episodes of a process and is gone thereafter.

Consequence: L3's protocol runs every variant in a fresh process, so all 200 L3 episodes are first-batch-cold, while flat cells are 80% warm. On the warm-up-matched comparison (flat cells restricted to their first batch, $n=20/\text{cell}$ — noisy but unbiased; `experiments/warmup_matched_f4.csv`), the object-axis premium largely dissolves at mid/high N: T1's L3 at $N=100$ scores 0.70 against L1/L2's matched 0.50, and T3's L3 tracks its L2 within noise from $N=25$ up. A genuine small-N premium remains (L3 is worst at $N \leq 25$ in all tasks).

Settling finding 4 properly needs warm L3 numbers: prepend one unscored burn-in batch per variant to the L3 eval protocol and re-run the 18 L3 cells (~10–12 h local, protocol addendum — not run unasked). The published warm-vs-cold asymmetry also means the study's L3 curves are conservative (true L3 performance is higher than reported).

Question 1

What is wrong with T2?

The inversion is statistically real, per episode

T2's two frozen eval sets share **bit-identical object poses** ($\max |\Delta| = 0.0$; the sampler draws all dims from one RNG stream even for fixed ranges), differing only in L2's added cabinet perturbation. So L1's eval is the same benchmark with an easier, training-matched cabinet — and L2 still wins:

N DEMOS	L1 SR	L2 SR	FISHER P	ODDS RATIO
10	0.12	0.17	0.42	1.50
25	0.22	0.11	0.056	0.44
50	0.19	0.35	0.016	2.30
100	0.30	0.32	0.88	1.10
200	0.25	0.29	0.63	1.23
400	0.23	0.41	0.0097	2.33
pooled $N \geq 50$	97/400	137/400	0.0024	1.63

Pose-paired McNemar (valid because poses match): only-L2 103 vs only-L1 63, $p=0.0024$. The inversion is T2-specific — T1 and T3 have $L1 \geq L2$ at every N. (*The table shows the originally recorded numbers, before the harness fix of Finding 0; on the clean re-eval the N=400 comparison is 0.37 vs 0.15 — the inversion survives and grows.*)

Mechanism: survivorship left L2 cleaner demos

- **Not the eval.** Same success criterion (drawer-body frame, identical params), same horizon (1200 steps vs ≤ 743 -step demos — 62% slack, censoring impossible), drawer closed at reset in both, no dead eval batches.
- **Not coverage or subregion luck.** L2 beats L1 in 11 of 12 pose quartiles — a broad win, not a trade. Cabinet dims do not modulate L2's success (all $p \geq 0.22$).
- **Demo quality differs, in L2's favor.** At matched geometry (L2 demos with cabinet within 2 cm of L1's fixed pose), L2's surviving demos are shorter and more direct: 672 steps / 3.27 m end-effector path vs L1's 694 / 3.40 m ($p \leq 0.0005$). The harsher filter (gen SR 30.6% vs 44.2%) kept the demonstrator's most efficient executions — better imitation targets in a compounding-error regime. A second, untested contributor: cabinet randomization as domain-randomization-style augmentation.
- **One-seed caveat.** With the training run as the noise unit, the inversion is only $\sim 1.3\sigma$ (paired t $p=0.24$), and within-level run noise is inversion-sized: L2's jump from N=25 (0.11) to N=50 (0.35) on nested demo subsets is $p=0.0001$. Episode-level significance assumes fixed policies; treat the inversion as real-but-fragile until a second seed exists.

Why T2's SR is low everywhere above L0

The prior finding stands after re-verification: effective epochs are ~ 18 for every T2 cell (verified exactly), yet L0 reaches 0.96 on the same budget, and losses converged — so not undertraining. Imitation loss is the study's lowest while rollouts fail: the signature of distribution shift compounding over a long, multi-phase horizon with success-only demos and no recovery behavior. The eval-horizon check was the one untested escape and it is now closed (min 457-step slack over the longest demo).

Why L2 > L1: policy-side, not eval-side

On the clean surface the inversion grows (L2 0.37 vs L1 0.15 at N=400; L2 above L1 at every $N \geq 25$). Diagnostic cross-evals attributed it to the policy rather than the eval sets (pooled episode-paired McNemar $p=0.0045$ for the policy effect vs $p=0.11$ for the set effect); per this report's scope they are excluded from all tables and archived under `results/`

diagnostics/ — matched policy/eval-set measurements only from here on. The mechanism consistent with all retained evidence: L2's harsher generation filter (69% vs 56% rejection) kept measurably shorter, more direct demonstrations — better imitation targets on a compounding-error task.

Stage funnels: T2's failure has an exact address



The same pick that runs at ≈ 0.95 in T1 at matched level and N fails 60–85% of the time here, ~ 300 steps into the rollout, after a drawer phase the policy handles fine (median ~ 155 steps to open, level-independent): compounding drift, localized to the mid-rollout grasp. The object axis (L3) makes that grasp harder still; the drawer never becomes the problem.

"Should the task be shortened?" As a diagnostic, yes: a variant starting with the drawer open isolates the real bottleneck and cuts $\sim 25\%$ of horizon. As a rescue, no: the funnel predicts its SR at roughly $P(\text{lift} \mid \text{open}) \approx 0.2\text{--}0.4$ at $N=400$ — the expensive part is the mid-rollout grasp under accumulated drift, not the drawer. The cheap decisive follow-ups now are a second seed for T2 L1/L2 (fragility caveat) and the 54-cell clean re-eval above.

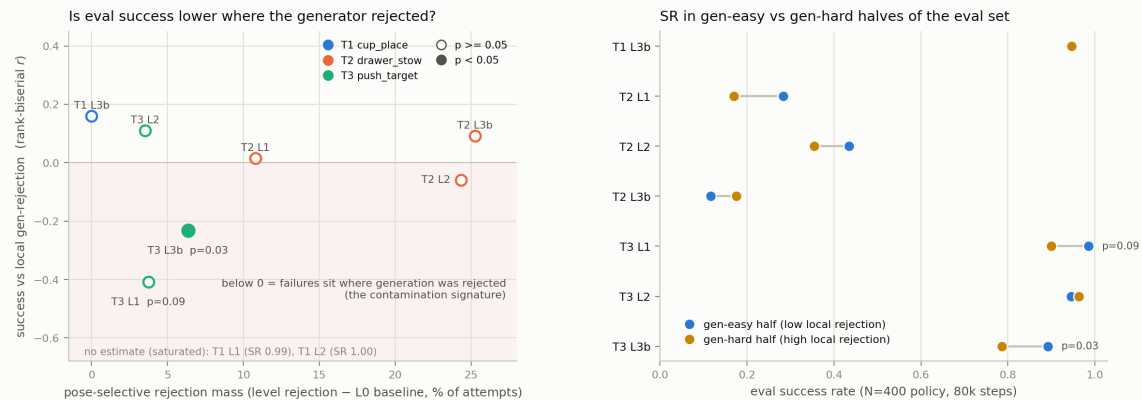
Question 2

Does generation success-filtering shape the measured SR?

Concern: generation only succeeds in a sub-distribution → demos exist only there → the policy succeeds only there → SR partly measures the generator's filter. **Method:** per cell, compare retained vs rejected generation-attempt initial states (KS per dim), then join the frozen eval set's per-episode initial states to outcomes and test whether success is higher where generation succeeded (rank-biserial, gen-easy/gen-hard split halves, nearest-training-demo distance using each cell's actual demo subset).

Generator filtering vs downstream eval success -- all randomized cells (L0 cells exempt: single fixed pose)

local gen-rejection = fraction rejected among an eval state's 25 nearest generation attempts (z-scored demo-observable dims).
T1 L1/L2 have ≤ 1 eval failure at N=400, so no effect is estimable there (shown as saturated).



The audit in one picture. Left: the contamination signature would be points deep in the shaded region (failures concentrated where generation was rejected) — no cell lands there significantly. Right: gen-easy vs gen-hard halves of each eval set differ by little, and in T2 L1/L2 the "hard" half isn't even the worse one.

CELL	REJECTION % (POSE-SELECTIVE)	STRONGEST FILTER SKEW	SUCCESS ↔ GEN-REJECTION	LINK	VERDICT
T1 L0	13.6 (baseline)	n/a — fixed pose	n/a		EXEMP
T1 L1	14.2 (≤ 0.6)	none ($p \geq 0.18$)	≤ 1 failure at N=400 — saturated		ABSEN
T1 L2	14.9 (≤ 1.3)	none ($p \geq 0.52$)	0 failures — saturated		ABSEN
T1 L3	13.4 (≈ 0)	none ($p \geq 0.23$)	null ($p = 0.37$)		ABSEN
	45.1 (baseline)	n/a — fixed pose	n/a		EXEMP

CELL	REJECTION % (POSE-SELECTIVE)	STRONGEST FILTER SKEW	SUCCESS ↔ GEN-REJECTION	LINK	VERDICT
T2 L0					
T2 L1	55.9 (≤10.8)	yaw , p=8e-15	null (p=0.92)		IMMAT
T2 L2	69.4 (≤24.3)	signed yaw, p=8e-20	null (p=0.62)		IMMAT
T2 L3	70.3 (≤25.2)	signed yaw, p=1e-22	null (p=0.44)		IMMAT
T3 L0	1.5 (baseline)	n/a — fixed pose	n/a		EXEMP
T3 L1	5.2 (≤3.7)	y, D=0.68 (all 22 rejects in one strip)	suggestive (p=0.09), bounded ≤5 pts		IMMAT
T3 L2	5.0 (≤3.5)	bearing, p=7e-5 (new — missed by gen_bias.csv)	null (p≥0.35)		IMMAT
T3 L3	7.8 (≤6.3)	bearing, p=6e-5	p=0.034, but anti-filter direction; demo-distance null		IMMAT

(L3 rows use the corrected L3b arms, reported as L3 per the naming decision. "Pose-selective" = level rejection minus the task's L0 baseline, since L0 rejects at a single fixed pose and is therefore pose-blind noise.)

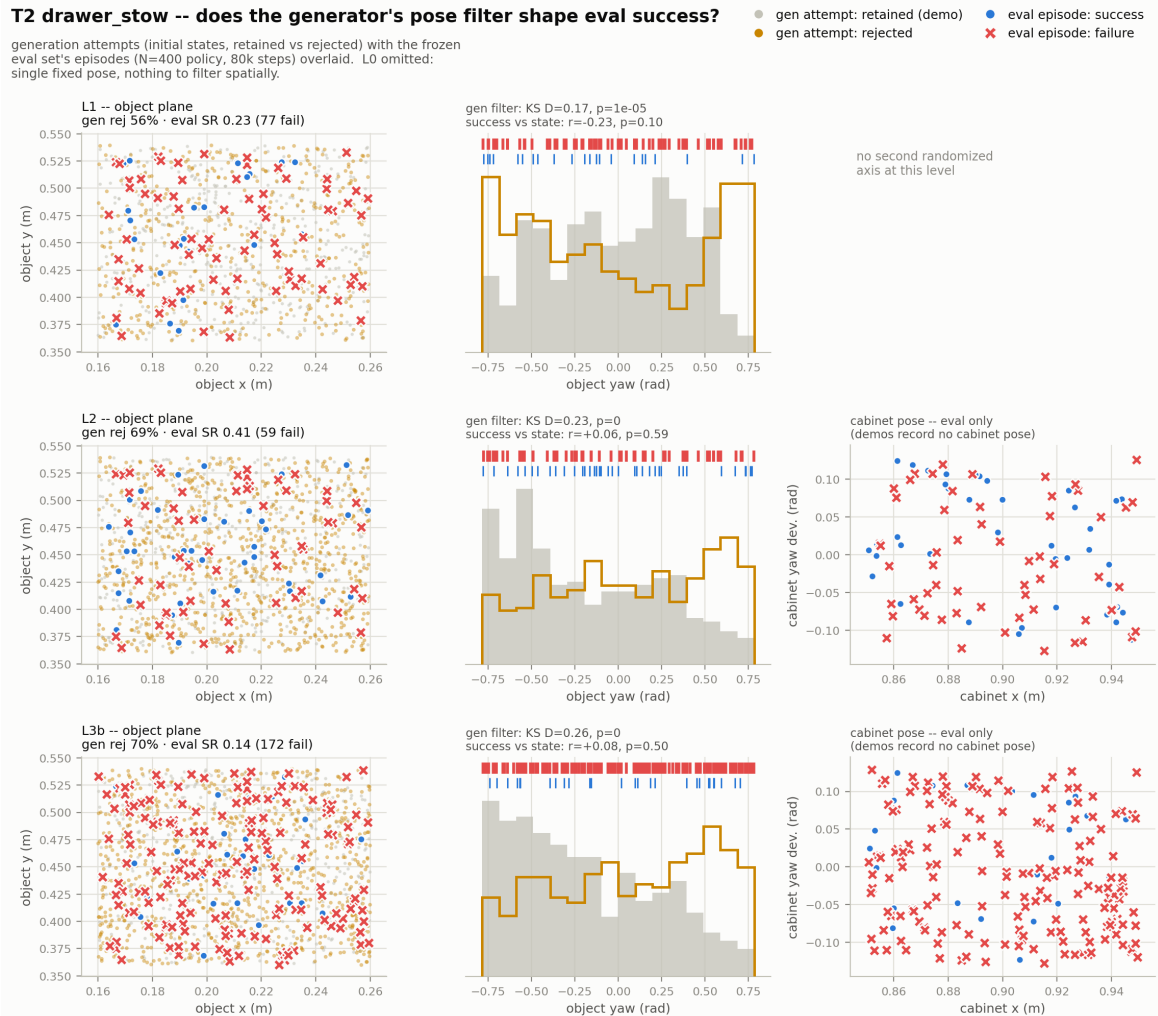
Per-task reading

- **T1 cup_place — clean.** ~14% rejection is pose-blind (L0 rejects the same fraction); no retained-vs-rejected skew anywhere.
- **T2 drawer_stow — huge, structured filter; zero downstream trace.** 56–70% rejection, genuinely yaw-selective. Yet success is flat across every filtered dim, and per-variant gen SR is uncorrelated with eval SR. T2's plateaus are a policy/training limit, not a coverage artifact — if anything the filter starved the +yaw region *without* hurting success there (anti-inflation).
- **T3 push_target — small, geometrically sharp filter; common-cause, not starvation.** The one $p < 0.01$ state-success effect (L3, object-x) runs in the *anti-filter* direction, and

nearest-demo distance is null: the low-bearing region is intrinsically hard for scripted generator and learned policy alike.

T2 drawer_stow -- does the generator's pose filter shape eval success?

generation attempts (initial states, retained vs rejected) with the frozen eval set's episodes (N=400 policy, 80k steps) overlaid. L0 omitted: single fixed pose, nothing to filter spatially.



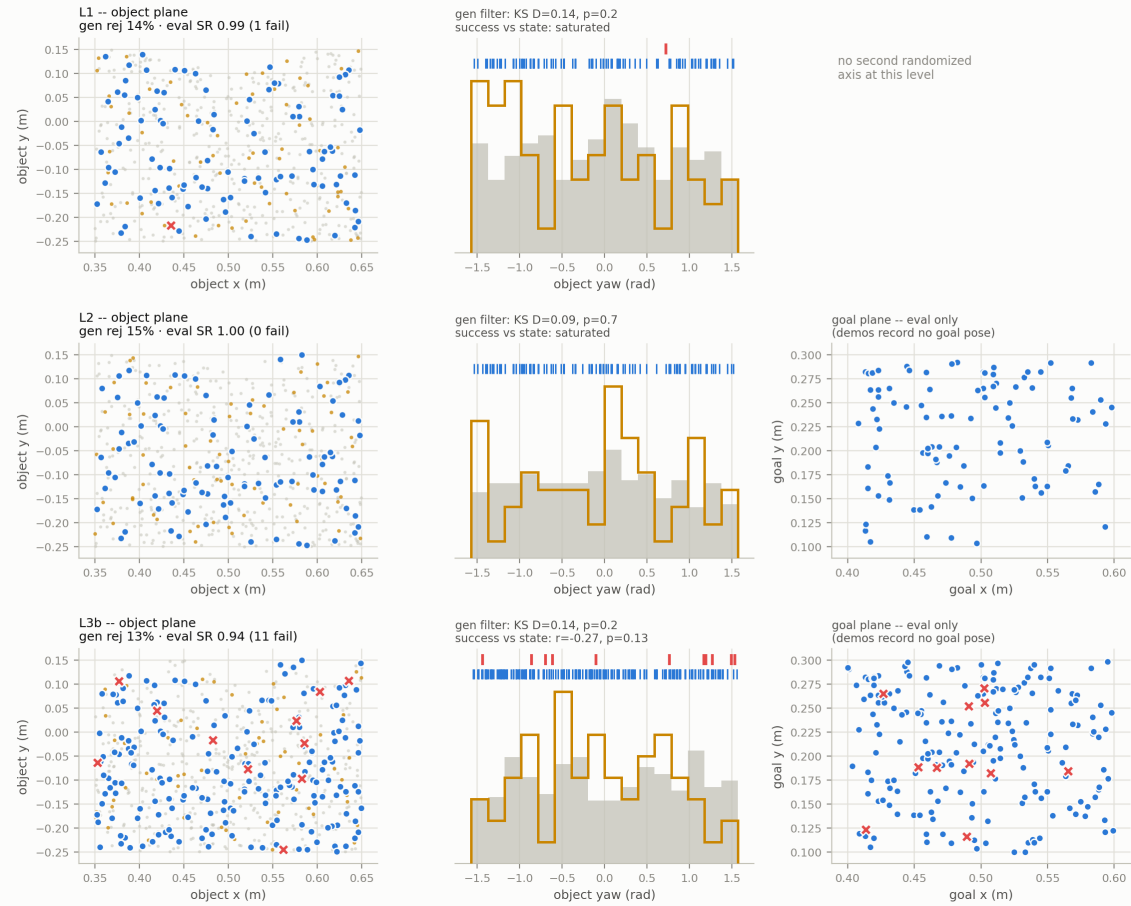
T2, the task where contamination was most plausible: the yaw filter is unmistakable (histograms), but eval failures (red x) fall everywhere — not where rejections (amber) concentrate.

T1 cup_place — per-level figure

T1 cup_place -- does the generator's pose filter shape eval success?

generation attempts (initial states, retained vs rejected) with the frozen eval set's episodes (N=400 policy, 80k steps) overlaid. L0 omitted: single fixed pose, nothing to filter spatially.

● gen attempt: retained (demo) ● eval episode: success
● gen attempt: rejected ✕ eval episode: failure

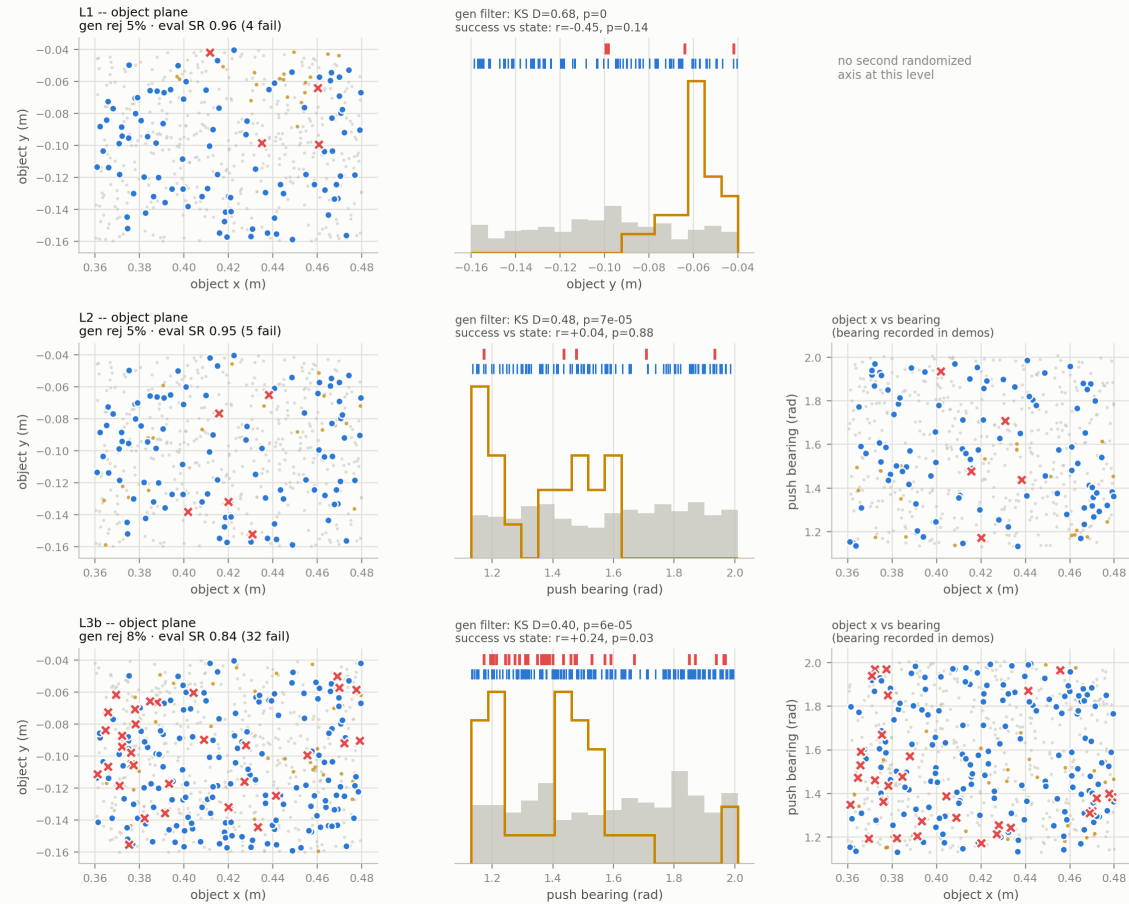


T3 push_target — per-level figure

T3 push_target -- does the generator's pose filter shape eval success?

generation attempts (initial states, retained vs rejected) with the frozen eval set's episodes (N=400 policy, 80k steps) overlaid. L0 omitted: single fixed pose, nothing to filter spatially.

● gen attempt: retained (demo) ● eval episode: success
● gen attempt: rejected ✗ eval episode: failure



Gotchas found during the audit (recorded in the journal)

- **L3 eval JSONs log batch=0 for all 200 outcomes.** The outcome list is variant-major; any per-episode L3 analysis must reindex `batch = index//20` or the join silently corrupts 90% of episodes (this artifact produced, then killed, a spurious $p=0.003$).
- **T1 demos record no goal pose; T2 demos no cabinet pose.** The generator's filter on those axes is unauditable end-to-end. T3 records its target, which is how the previously-missed bearing filter surfaced.
- **Power limits:** T1 L1/L2 and T3 L0–L2 have ≤ 6 failures at N=400 (near-zero power; N=100 policies used as secondary — still null).

Provenance

branch `results-analysis` (worktree `.claude/worktrees/results-analysis`),
2026-08-21 → 08-22

clean sweep: `results/eval_*_080000_fixed.json` (54 flat + D24 pair) ·
`results/diagnostics/eval_T2_L3b_n*_080000_fixed.json` (6, stage-
instrumented) – 62 evals, 0 failures, ~18 h managed local GPU

final tables: `experiments/clean_surface.csv` · `clean_nstar.csv` ·
`t2_stage_funnel_full.csv` (deprecated: `stale_corrected_sr.csv`,
`batch0_corrected_sr.csv`)

earlier stats: `experiments/t2_l1_vs_l2_tests.csv` · `t2_demo_quality.csv` ·
`t2_pose_quartile_deltas.csv` · `genbias_link.csv` · `t2_followup.csv`

figures: `paper/figures/genbias_link_{summary,T1,T2,T3}.png` ·
`corrected_vs_published_sr.png` (clean surface) · `t2_stage_funnel.png`

cross-evals (mismatched policy/eval-set) excluded from this report and
archived: `results/diagnostics/eval_T2_xeval_*.json`

harness fix: `src/cog/eval/rollout_eval.py` (t=0 guard + --stages) ·

orchestrators `scripts/ops/resweep_eval.py`, `resweep_l3_phase.py`

journal: `docs/journal.md` entries 2026-08-21/22 (bug, revision, sweep, tmux
prefix-match gotcha) · `docs/running_jobs.md` + `timings.md` updated